

Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
Institut Teknologi Sepuluh Nopember

Laporan Akhir Praktikum Jaringan Komputer

Firewall dan NAT

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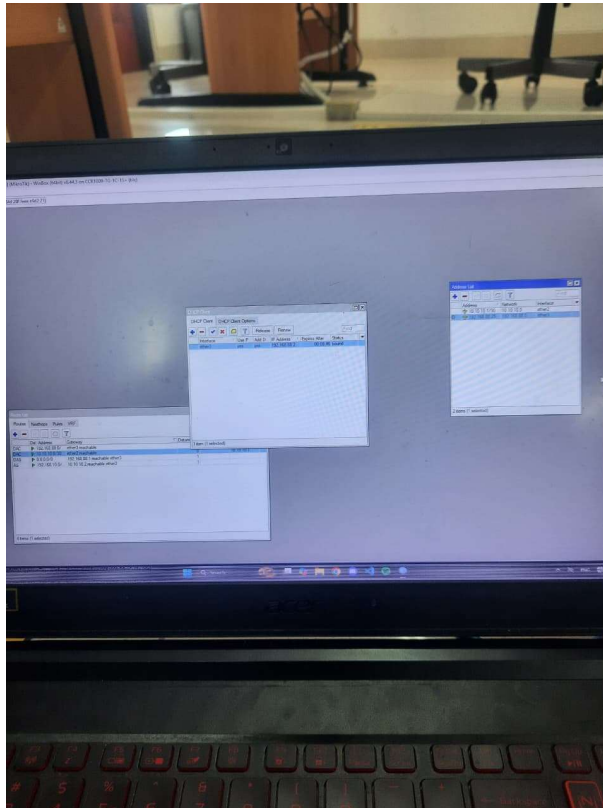
03/06/2026

1 Langkah-Langkah Percobaan

Terdapat tiga bagian utama dalam percobaan Modul 4 ini, yaitu Firewall NAT Dasar, Firewall Filter Input dan Forward, serta Firewall NAT Forwarding.

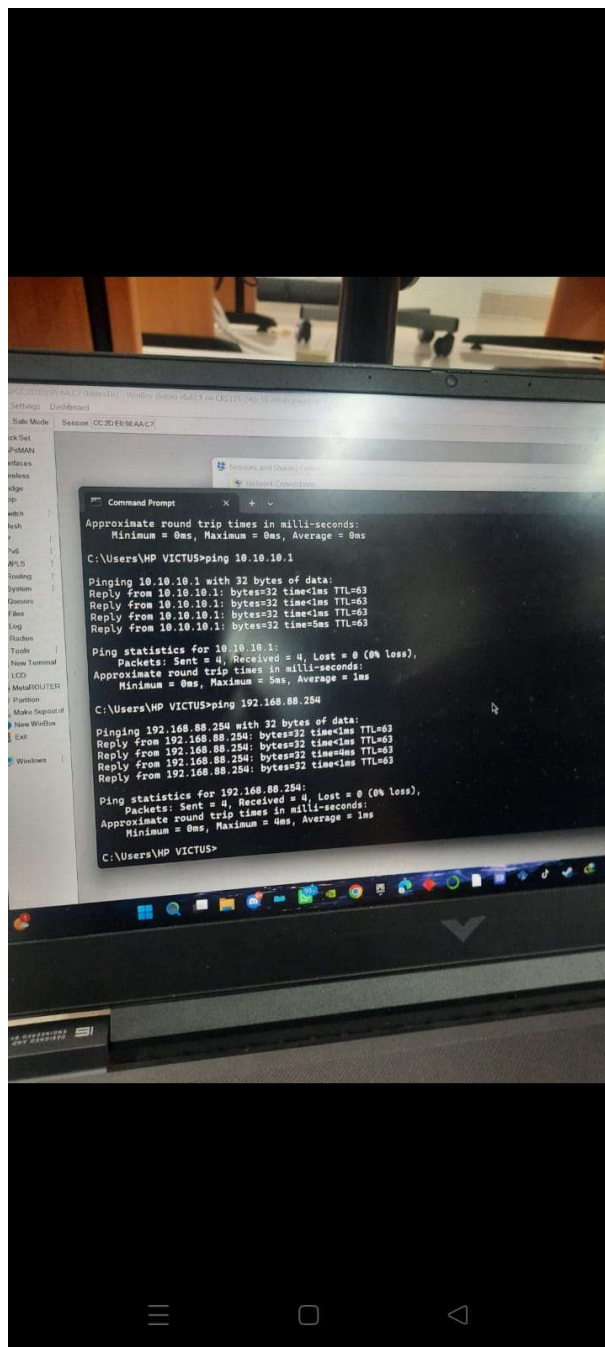
1.1 Praktikum 1: Firewall NAT Dasar

1. Membuat topologi Praktikum 1 pada PNETLab sesuai dengan modul.
2. Melakukan konfigurasi awal pada Router A.



Gambar 1: Konfigurasi awal Router A

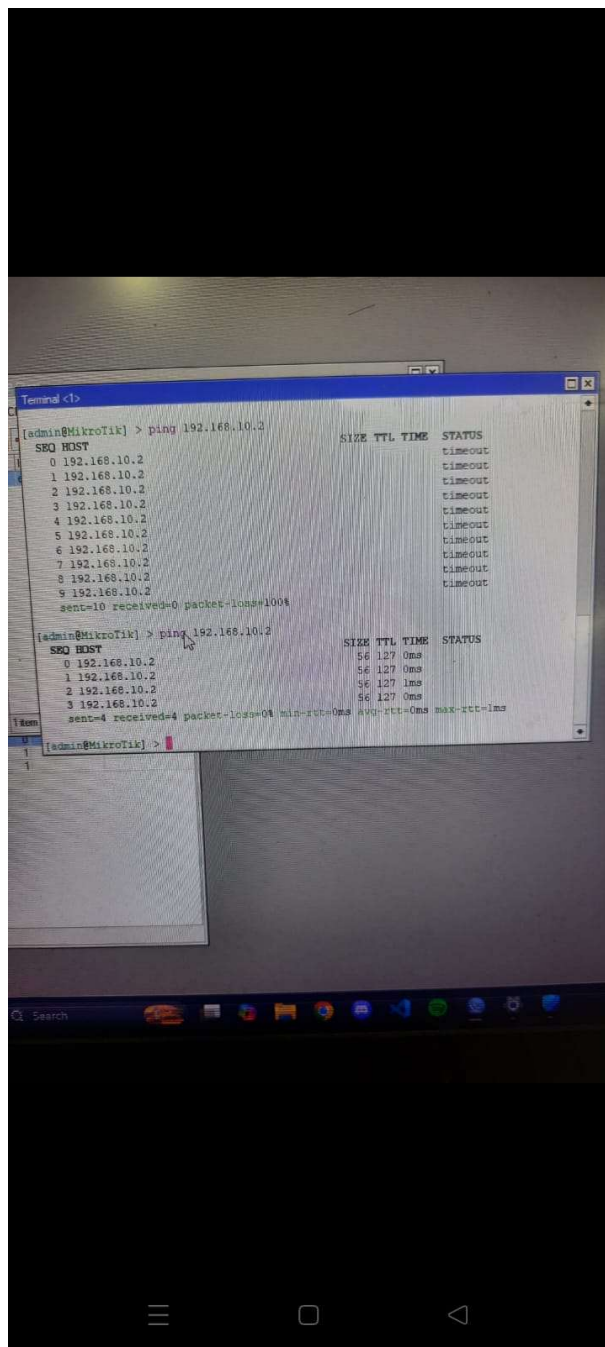
3. Melakukan konfigurasi IP address pada Router B.
4. Menambahkan static route agar router dapat saling terhubung.
5. Mengaktifkan NAT masquerade agar jaringan lokal dapat mengakses jaringan luar.
6. Melakukan pengujian koneksi menggunakan perintah ping dari Router B.



Gambar 2: Pengujian koneksi dari Router B

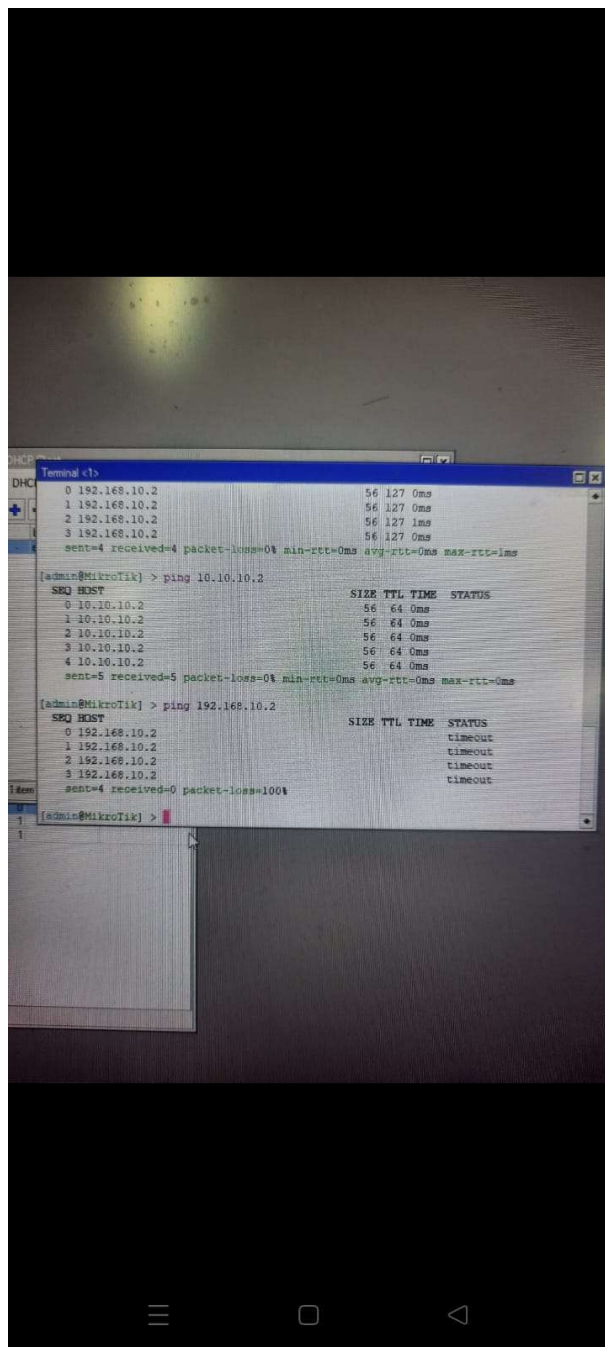
1.2 Praktikum 2: Firewall Filter Input dan Forward

1. Menggunakan topologi yang sudah dibuat untuk melakukan pengujian firewall filter.
2. Menambahkan rule firewall dengan chain *input*.
3. Melakukan pengujian ping untuk melihat hasil dari rule chain *input*.



Gambar 3: Pengujian firewall chain input

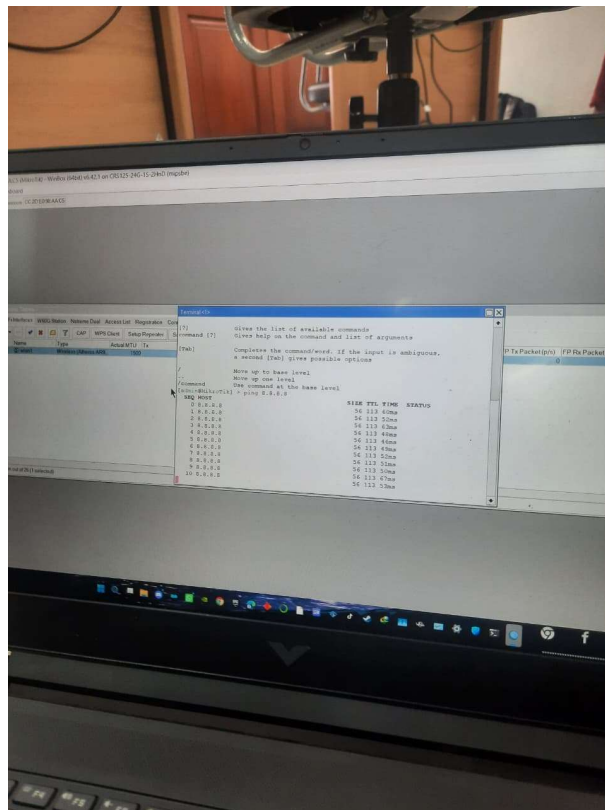
4. Menonaktifkan rule chain *input*.
5. Membuat rule firewall baru dengan chain *forward*.
6. Melakukan pengujian ping untuk melihat hasil dari rule chain *forward*.



Gambar 4: Pengujian firewall chain forward

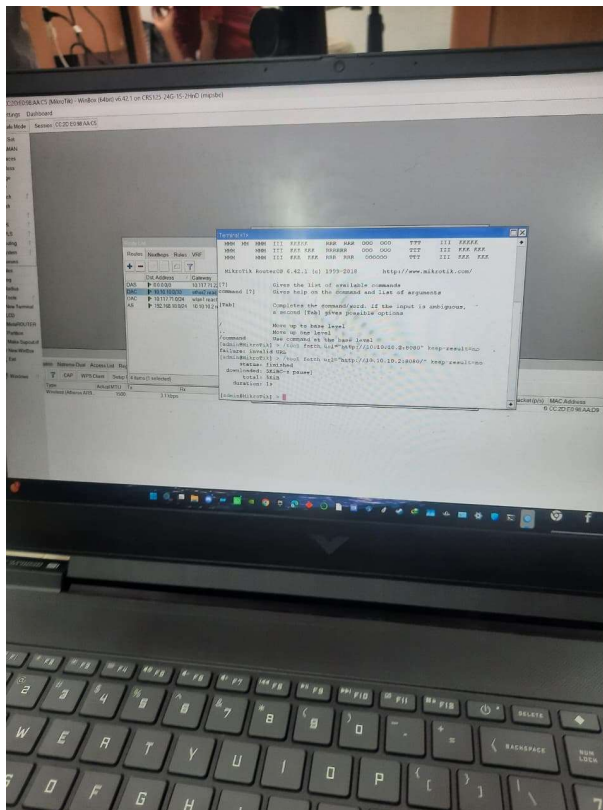
1.3 Praktikum 3: Firewall NAT Forwarding

1. Menyiapkan topologi untuk percobaan port forwarding.
2. Menyiapkan PC client sebagai web server.
3. Melakukan konfigurasi NAT forwarding pada router.



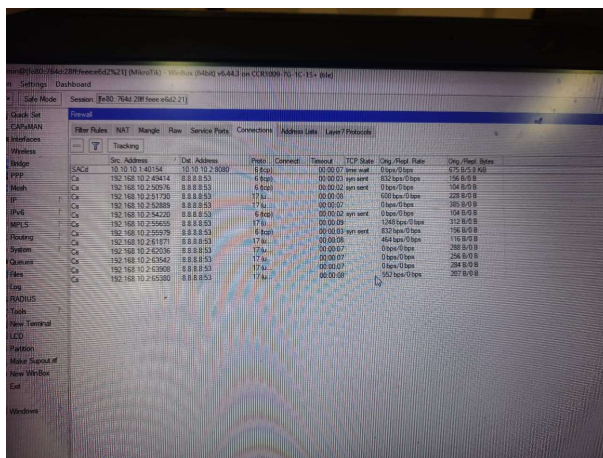
Gambar 5: Persiapan jaringan dan web server

4. Melakukan pengujian akses web server menggunakan fetch tool.



Gambar 6: Pengujian port forwarding menggunakan fetch tool

5. Mengecek hasil koneksi pada connection tracking.
6. Jika koneksi berhasil tercatat, maka konfigurasi NAT forwarding sudah berjalan.



Gambar 7: Hasil connection tracking

2 Analisis Hasil Percobaan

Pada Praktikum 1, konfigurasi Firewall NAT Dasar berhasil dilakukan. Router A dan Router B dapat saling terhubung, dan Router B dapat menjadi gateway untuk jaringan lokal. NAT masquerade

digunakan agar client pada jaringan lokal dapat mengakses jaringan luar menggunakan alamat IP router.

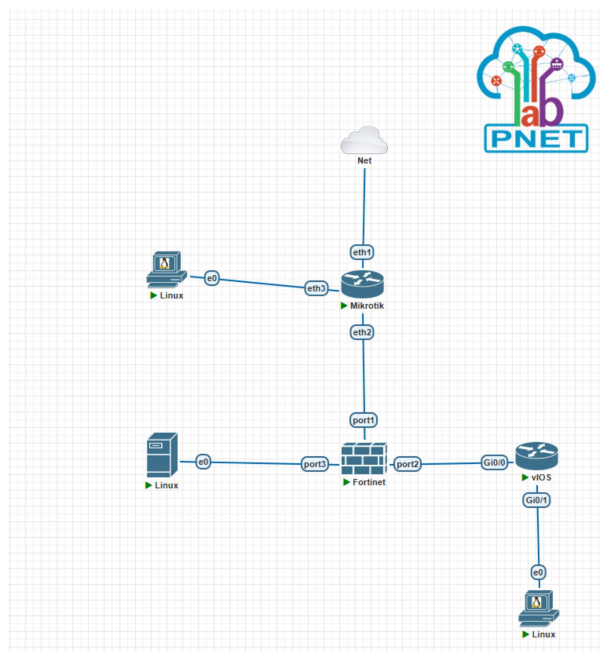
Pada Praktikum 2, perbedaan chain *input* dan chain *forward* dapat terlihat dari hasil pengujian. Chain *input* digunakan untuk memfilter paket yang menuju router secara langsung, sedangkan chain *forward* digunakan untuk memfilter paket yang melewati router menuju perangkat lain.

Pada Praktikum 3, NAT forwarding berhasil digunakan untuk meneruskan akses dari jaringan luar menuju web server yang berada di jaringan lokal. Hal ini menunjukkan bahwa web server lokal dapat diakses dari luar jika terdapat rule NAT forwarding yang sesuai.

3 Hasil Tugas Modul

Pada tugas modul ini dibuat topologi jaringan yang terdiri dari WAN, LAN, dan DMZ. Perangkat yang digunakan adalah MikroTik ISP, FortiGate, Cisco Router, Client LAN TinyCore Linux, Client WAN TinyCore Linux, dan Ubuntu Server sebagai server DMZ.

3.1 Topologi Tugas Modul



Gambar 8: Topologi jaringan tugas modul

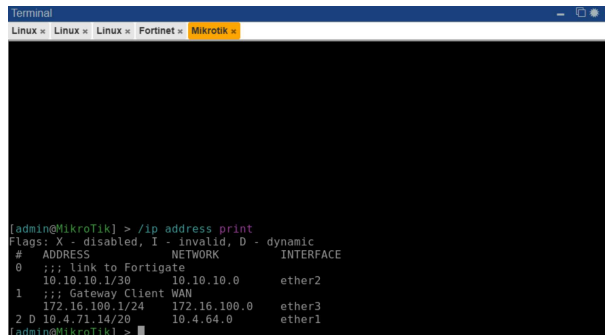
3.2 IP Address Tugas Modul

1. MikroTik ISP ether1 menggunakan DHCP Client dari jaringan lab.
2. MikroTik ISP ether2 menggunakan IP 10.10.10.1/30.
3. MikroTik ISP ether3 menggunakan IP 172.16.100.1/24.
4. FortiGate port1 menggunakan IP 10.10.10.2/30.
5. FortiGate port2 menggunakan IP 10.20.20.1/30.

6. FortiGate port3 menggunakan IP 192.168.20.1/24.
7. Cisco Router G0/0 menggunakan IP 10.20.20.2/30.
8. Cisco Router G0/1 menggunakan IP 192.168.10.1/24.
9. Client LAN menggunakan IP 192.168.10.10/24 dengan gateway 192.168.10.1.
10. Client WAN menggunakan IP 172.16.100.10/24 dengan gateway 172.16.100.1.
11. Ubuntu Server DMZ menggunakan IP 192.168.20.10/24 dengan gateway 192.168.20.1.

3.3 Konfigurasi MikroTik ISP

1. Mengatur interface ether1 sebagai DHCP Client.
2. Mengatur IP address ether2 sebagai link menuju FortiGate.
3. Mengatur IP address ether3 sebagai gateway Client WAN.

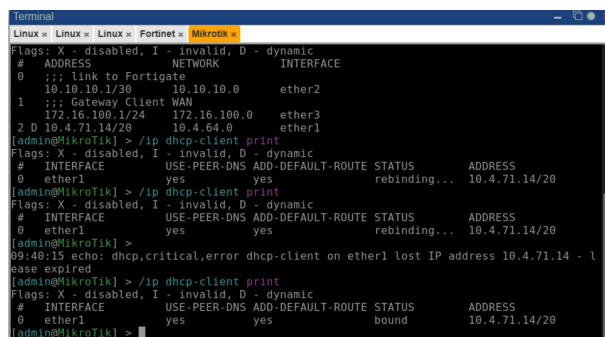


```

[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; link to Fortigate
10.10.10.1/30 10.10.10.0 ether2
1 ;; Gateway Client WAN
172.16.100.1/24 172.16.100.0 ether3
2 D 10.4.71.14/20 10.4.64.0 ether1
[admin@MikroTik] >

```

Gambar 9: Konfigurasi IP address MikroTik



```

[admin@MikroTik] > /ip dhcp-client print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes rebinding... 10.4.71.14/20
[admin@MikroTik] > /ip dhcp-client print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes rebinding... 10.4.71.14/20
[admin@MikroTik] >
09:40:15 echo: dhcp,critical,error dhcp-client on ether1 lost IP address 10.4.71.14 - l
ease expired
[admin@MikroTik] > /ip dhcp-client print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes bound 10.4.71.14/20
[admin@MikroTik] >

```

Gambar 10: Konfigurasi DHCP Client MikroTik

4. Mengatur NAT masquerade pada MikroTik.
5. Menambahkan static route menuju jaringan LAN dan DMZ.
6. Melakukan ping dari MikroTik ke FortiGate.

```

Terminal
Linux x Linux x Linux x Fortinet x Mikrotik x
1 ::: Gateway Client WAN
172.16.100.1/24 172.16.100.0 ether3
2 D 10.4.71.14/20 10.4.64.0 ether1
[admin@Mikrotik] > /ip dhcp-client print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes rebinding... 10.4.71.14/20
[admin@Mikrotik] > /ip dhcp-client print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes rebinding... 10.4.71.14/20
[admin@Mikrotik] >
09:40:15 echo: dhcp,critical,error dhcp-client on ether1 lost IP address 10.4.71.14 - l
ease expired
[admin@Mikrotik] > /ip dhcp-client print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes bound 10.4.71.14/20
[admin@Mikrotik] > /ip firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
0 ::: NAT to Internet
chain=srcnat action=masquerade out-interface=ether1
[admin@Mikrotik] >

```

Gambar 11: Konfigurasi NAT MikroTik

```

Terminal
Linux x Linux x Linux x Fortinet x Mikrotik x
lease expired
[admin@Mikrotik] > /ip dhcp-client print
Flags: X - disabled, I - invalid, D - dynamic
# INTERFACE USE-PEER-DNS ADD-DEFAULT-ROUTE STATUS ADDRESS
0 ether1 yes yes bound 10.4.71.14/20
[admin@Mikrotik] > /ip firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
0 ::: NAT to Internet
chain=srcnat action=masquerade out-interface=ether1
[admin@Mikrotik] > /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.64.1 1
1 ADC 10.4.64.0/20 10.4.71.14 ether1 0
2 ADC 10.10.10.0/30 10.10.10.1 ether2 0
3 ADC 172.16.100.0/24 172.16.100.1 ether3 0
4 A S ::: Route to LAN via Fortigate
192.168.10.0/24 10.10.10.2 1
5 A S ::: Route to DMZ via fortigate
192.168.20.0/24 10.10.10.2 1
[admin@Mikrotik] >

```

Gambar 12: Konfigurasi routing MikroTik

```

Terminal
Linux x Linux x Linux x Fortinet x Mikrotik x
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.64.1 1
1 ADC 10.4.64.0/20 10.4.71.14 ether1 0
2 ADC 10.10.10.0/30 10.10.10.1 ether2 0
3 ADC 172.16.100.0/24 172.16.100.1 ether3 0
4 A S ::: Route to LAN via Fortigate
192.168.10.0/24 10.10.10.2 1
5 A S ::: Route to DMZ via fortigate
192.168.20.0/24 10.10.10.2 1
[admin@Mikrotik] > ping 10.10.10.2
SEQ HOST SIZE TTL TIME STATUS
0 10.10.10.2 56 255 0ms
1 10.10.10.2 56 255 1ms
2 10.10.10.2 56 255 2ms
3 10.10.10.2 56 255 1ms
4 10.10.10.2 56 255 0ms
5 10.10.10.2 56 255 1ms
sent=6 received=6 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=2ms
[admin@Mikrotik] >

```

Gambar 13: Pengujian ping MikroTik ke FortiGate

3.4 Konfigurasi FortiGate

1. Mengatur port1 sebagai interface WAN.
2. Mengatur port2 sebagai interface menuju Cisco Router.
3. Mengatur port3 sebagai interface DMZ.

```

Terminal
Linux x Linux x Fortinet x Mikrotik x
Password:
Welcome!
FortiGate-VM64-KVM # show system interface port1
config system interface
  edit "port1"
    set vdom "root"
    set ip 10.10.10.2 255.255.255.252
    set allowaccess ping https ssh http
    set type physical
    set alias "WAN_TO_MIKROTIK"
    set snmp-index 1
  next
end
FortiGate-VM64-KVM # show system interface port2
config system interface
  edit "port2"
    set vdom "root"
    set ip 10.20.20.1 255.255.255.252
    set allowaccess ping https ssh http
    set type physical
    set alias "INSIDE_to_cisco"
    set snmp-index 2
  next
end
FortiGate-VM64-KVM # show system interface port3
config system interface
  edit "port3"
    set vdom "root"
    set ip 192.168.20.1 255.255.255.0
    set allowaccess ping https ssh http
    set type physical
    set alias "DMZ_to_Server"
    set snmp-index 3
  next
end
FortiGate-VM64-KVM #

```

Gambar 14: Konfigurasi interface FortiGate

4. Menambahkan default route menuju MikroTik ISP.
5. Menambahkan route menuju jaringan LAN melalui Cisco Router.
6. Membuat firewall policy untuk mengatur akses antar jaringan.
7. Membuat VIP agar Client WAN dapat mengakses web server DMZ.
8. Melakukan ping dari FortiGate ke server DMZ.

```

Terminal
Linux x Linux x Fortinet x Mikrotik x
edit "port2"
  set vdom "root"
  set ip 10.20.20.1 255.255.255.252
  set allowaccess ping https ssh http
  set type physical
  set alias "INSIDE_to_cisco"
  set snmp-index 2
next
end
FortiGate-VM64-KVM # show system interface port3
config system interface
  edit "port3"
    set vdom "root"
    set ip 192.168.20.1 255.255.255.0
    set allowaccess ping https ssh http
    set type physical
    set alias "DMZ_to_Server"
    set snmp-index 3
  next
end
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       0 - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default
Routing table for VRF=0
S*  0.0.0.0/0 [10/0] via 10.10.10.1, port1, [1/0]
C   10.10.10.0/30 is directly connected, port1
C   10.20.20.0/30 is directly connected, port2
C   192.168.10.0/24 [10/0] via 10.20.20.2, port2, [1/0]
C   192.168.20.0/24 is directly connected, port3
FortiGate-VM64-KVM #

```

Gambar 15: Konfigurasi routing FortiGate

```

Linux < Linux < Linux < Fortinet < Mikrotik <
edit 1
set name "LAN to WAN"
set uuid 7a2077a4-5f1f-51f1-9591-45cd705ad6da
set srcintf "port2"
set dstintf "port1"
set action accept
set srcaddr "LAN_192.168.10.0"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next
edit 2
set name "LAN to DMZ"
set uuid b9d53114-5f1f-51f1-33a8-af1406cd211d
set srcintf "port2"
set dstintf "port3"
set action accept
set srcaddr "LAN_192.168.10.0"
set dstaddr "DMZ_Server_192.168.20.10"
set schedule "always"
set service "ALL"
next
edit 3
set name "WAN to DMZ HTTP"
set uuid 212f7ab8-5f20-51f1-b3d4-a79d47af7509
set srcintf "port1"
set dstintf "port3"
set action accept
set srcaddr "Client_WAN_172.16.100.10"
set dstaddr "VIP_HTTP_DMZ"
set schedule "always"
set service "HTTP"
next
edit 4
set name "DMZ to WAN"
set uuid 645afd64-5f23-51f1-8391-9c525aae36da
set srcintf "port3"
set dstintf "port1"
set action accept
set srcaddr "DMZ_Server_192.168.20.10"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next

```

Gambar 16: Konfigurasi firewall policy FortiGate

```

Linux < Linux < Linux < Fortinet < Mikrotik <
set service "ALL"
next
edit 3
set name "WAN to DMZ HTTP"
set uuid 212f7ab8-5f20-51f1-b3d4-a79d47af7509
set srcintf "port1"
set dstintf "port3"
set action accept
set srcaddr "Client_WAN_172.16.100.10"
set dstaddr "VIP_HTTP_DMZ"
set schedule "always"
set service "HTTP"
next
edit 4
set name "DMZ to WAN"
set uuid 645afd64-5f23-51f1-8391-9c525aae36da
set srcintf "port3"
set dstintf "port1"
set action accept
set srcaddr "DMZ_Server_192.168.20.10"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next
end
FortiGate-VM64-KVM # FortiGate-VM64-KVM #
FortiGate-VM64-KVM # FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM # show firewall vip
config firewall vip
edit "VIP_HTTP_DMZ"
set uuid f90c6cea-5f1f-51f1-bd58-012bedbffa5
set extip 10.10.10.2
set mappedip "192.168.20.10"
set extintf "port1"
set portforward enable
set extport 80
set mappedport 80
next
end
FortiGate-VM64-KVM #

```

Gambar 17: Konfigurasi VIP FortiGate

```
Linux x Linux x Linux x Fortinet x Mikrotik x
next
edit 4
set name "DMZ to WAN"
set uuid 645af6d4-5f23-51f1-8391-9c525aee36da
set srcintf "port3"
set dstintf "port1"
set action accept
set srcaddr "DMZ_Server_192.168.20.10"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next
end
FortiGate-VM64-KVM # FortiGate-VM64-KVM #
FortiGate-VM64-KVM # FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM #
FortiGate-VM64-KVM # show firewall vip
config firewall vip
edit "VIP_HTTP_DMZ"
set uuid f99c6cea-5f1f-51f1-bd58-012bedbffa5
set extip 10.10.10.2
set mappedip "192.168.20.10"
set extintf "port1"
set portforward enable
set extport 80
set mappedport 80
next
end
FortiGate-VM64-KVM # execute ping 192.168.20.10
PING 192.168.20.10 (192.168.20.10): 56 data bytes
64 bytes from 192.168.20.10: icmp_seq=0 ttl=64 time=0.8 ms
64 bytes from 192.168.20.10: icmp_seq=1 ttl=64 time=0.4 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=64 time=0.5 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=64 time=1.6 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=64 time=1.0 ms
... 192.168.20.10 ping statistics ...
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.4/0.8/1.6 ms
FortiGate-VM64-KVM # D
```

Gambar 18: Pengujian ping FortiGate ke server DMZ

3.5 Konfigurasi Cisco Router

1. Mengatur interface G0/0 menuju FortiGate.
2. Mengatur interface G0/1 sebagai gateway LAN.
3. Menambahkan default route menuju FortiGate.

```
Terminal
vIOS x
*Jun  3 10:03:19.248: %PLATFORM-5-SIGNATURE VERIFIED: Image "flash0:/vios-adventerprise
k9-m" passed code signing verification*****
*****
* IOSv is strictly limited to use for evaluation, demonstration and IOS *
* education. IOSv is provided as-is and is not supported by Cisco's *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****
Cisco-LAN>
Cisco-LAN>
Cisco-LAN>
Cisco-LAN>
Cisco-LAN>
Cisco-LAN>show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.20.20.2      YES NVRAM  up          up
GigabitEthernet0/1  192.168.10.1    YES NVRAM  up          up
GigabitEthernet0/2  unassigned      YES NVRAM  administratively down down
GigabitEthernet0/3  unassigned      YES NVRAM  administratively down down
Cisco-LAN>
```

Gambar 19: Konfigurasi interface Cisco Router

```

Terminal
vIOS -
GigabitEthernet0/2      unassigned      YES NVRAM   administratively down down
GigabitEthernet0/3      unassigned      YES NVRAM   administratively down down
Cisco-LAN>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is 10.20.20.1 to network 0.0.0.0

S*  0.0.0.0/0 [1/0] via 10.20.20.1
C    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    10.20.20.0/30 is directly connected, GigabitEthernet0/0
L    10.20.20.2/32 is directly connected, GigabitEthernet0/0
C    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.10.0/24 is directly connected, GigabitEthernet0/1
L    192.168.10.1/32 is directly connected, GigabitEthernet0/1
Cisco-LAN>

```

Gambar 20: Konfigurasi routing Cisco Router

4. Melakukan ping dari Cisco Router ke FortiGate.
5. Melakukan ping dari Cisco Router ke server DMZ.

```

Terminal
vIOS -
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is 10.20.20.1 to network 0.0.0.0

S*  0.0.0.0/0 [1/0] via 10.20.20.1
C    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    10.20.20.0/30 is directly connected, GigabitEthernet0/0
L    10.20.20.2/32 is directly connected, GigabitEthernet0/0
C    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.10.0/24 is directly connected, GigabitEthernet0/1
L    192.168.10.1/32 is directly connected, GigabitEthernet0/1
Cisco-LAN>ping 10.20.20.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Cisco-LAN>

```

Gambar 21: Pengujian ping Cisco Router ke FortiGate

```

Terminal
vIOS - Linux - Linux - Fortinet -
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Cisco-LAN>ping 192.168.20.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
Cisco-LAN>
*Jun  3 10:06:41.450: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
*Jun  3 10:06:42.448: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
*Jun  3 10:06:46.445: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Jun  3 10:06:47.449: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Cisco-LAN>ping 192.168.20.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
Cisco-LAN>

```

Gambar 22: Pengujian ping Cisco Router ke server DMZ

3.6 Konfigurasi Ubuntu Server DMZ

1. Mengatur IP address Ubuntu Server DMZ.
2. Mengatur default gateway menuju FortiGate.
3. Mengecek IP address dan routing table.

```

Aug 03 18:02:46 km systemd[1]: Starting a high performance web server and a reverse proxy server...
Aug 03 18:02:47 km systemd[1]: Started a high performance web server and a reverse proxy server.
root@dmz:~#
1: 192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
2: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
3: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
4: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
5: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
6: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
7: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
8: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
9: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
10: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
11: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
12: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
13: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
14: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
15: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
16: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
17: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
18: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
19: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
20: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
21: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
22: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
23: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
24: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
25: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
26: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
27: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
28: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
29: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
30: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
31: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
32: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
33: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
34: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
35: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
36: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
37: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
38: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
39: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
40: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
41: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
42: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
43: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
44: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
45: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
46: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
47: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
48: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
49: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
50: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
51: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
52: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
53: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
54: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
55: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
56: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
57: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
58: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
59: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
60: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
61: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
62: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
63: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
64: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
65: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
66: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
67: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
68: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
69: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
70: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
71: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
72: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
73: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
74: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
75: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
76: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
77: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
78: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
79: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
80: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
81: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
82: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
83: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
84: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
85: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
86: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
87: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
88: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
89: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
90: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
91: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
92: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
93: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
94: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
95: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
96: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
97: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
98: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
99: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1
100: eth0:192.168.20.1/24 dev eth0 proto static scope link src 192.168.20.1

```

Gambar 23: Konfigurasi IP Ubuntu Server DMZ

```

root@dmz:~# ip route
valid:111 forever preferred:111 forever
default via 192.168.20.1 dev eth0 proto static
192.168.20.0/24 dev eth0 proto kernel scope link src 192.168.20.1
root@dmz:~#

```

Gambar 24: Routing table Ubuntu Server DMZ

4. Menginstal dan menjalankan Nginx.
5. Mengubah halaman web menjadi teks Tumod 4 DMZ Firewall kelompok 22.
6. Melakukan pengujian web server dari Ubuntu Server DMZ.

```

root@dmz:~# systemctl status nginx
nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/lib/systemd/system/nginx.service; enabled; vendor preset: enabled)
Active: active (running) since Wed 2022-06-03 10:02:47 CST; 10min ago
Process: 598 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on: master process on: (code=exited, status=0/SUCCESS)
Process: 602 ExecStart=/usr/sbin/nginx -g daemon on: master process on: (code=exited, status=0/SUCCESS)
Main PID: 603 (nginx)
Tasks: 5 (limit: 4620)
Memory: 14.1M
Group: nginx
CGroup: /system.slice/nginx.service
├─603 nginx: master process /usr/sbin/nginx -g daemon on: master process on:
└─604 nginx: worker process

```

Gambar 25: Status Nginx pada Ubuntu Server DMZ

```

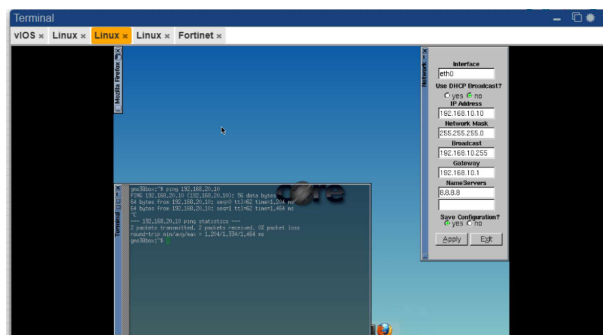
192.168.20.0/24 dev eth0 proto kernel scope link src 192.168.20.1
root@dmz:~# curl http://localhost
Tumod 4 DMZ Firewall kelompok 22
root@dmz:~#

```

Gambar 26: Pengujian web server dari Ubuntu Server DMZ

3.7 Konfigurasi dan Pengujian Client LAN

1. Mengatur IP Client LAN dengan IP 192.168.10.10/24.
2. Mengatur gateway Client LAN ke 192.168.10.1.



Gambar 27: Konfigurasi IP Client LAN

3. Melakukan ping dari Client LAN ke gateway.
4. Melakukan ping dari Client LAN ke server DMZ.
5. Melakukan akses web server DMZ dari Client LAN.


```

round-trip min/avg/max = 1.204/1.334/1.454 ms
gns3@box1:~$ ping 192.168.10.1
PING 192.168.10.1 (192.168.10.1): 56 data bytes
64 bytes from 192.168.10.1: seq=0 ttl=255 time=0.950 ms
64 bytes from 192.168.10.1: seq=1 ttl=255 time=1.215 ms
64 bytes from 192.168.10.1: seq=2 ttl=255 time=1.877 ms
64 bytes from 192.168.10.1: seq=3 ttl=255 time=2.096 ms
64 bytes from 192.168.10.1: seq=4 ttl=255 time=2.879 ms
64 bytes from 192.168.10.1: seq=5 ttl=255 time=1.952 ms
^C
--- 192.168.10.1 ping statistics ---
6 packets transmitted, 6 packets received, 0% packet loss
round-trip min/avg/max = 0.950/1.818/2.879 ms
gns3@box1:~$

```

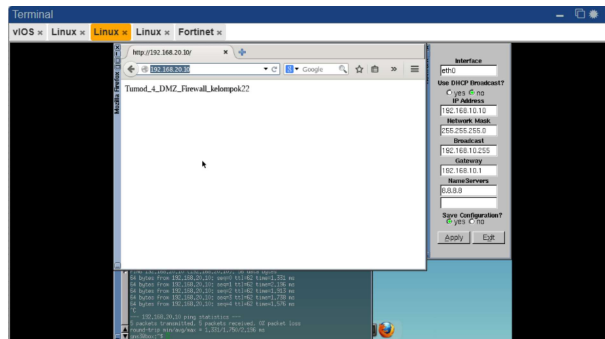
Gambar 28: Pengujian ping Client LAN ke gateway

```

gns3@box1:~$ ping 192.168.20.10
PING 192.168.20.10 (192.168.20.10): 56 data bytes
64 bytes from 192.168.20.10: seq=0 ttl=62 time=1.331 ms
64 bytes from 192.168.20.10: seq=1 ttl=62 time=2.196 ms
64 bytes from 192.168.20.10: seq=2 ttl=62 time=1.913 ms
64 bytes from 192.168.20.10: seq=3 ttl=62 time=1.738 ms
64 bytes from 192.168.20.10: seq=4 ttl=62 time=1.576 ms
^C
--- 192.168.20.10 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.331/1.750/2.196 ms
gns3@box1:~$

```

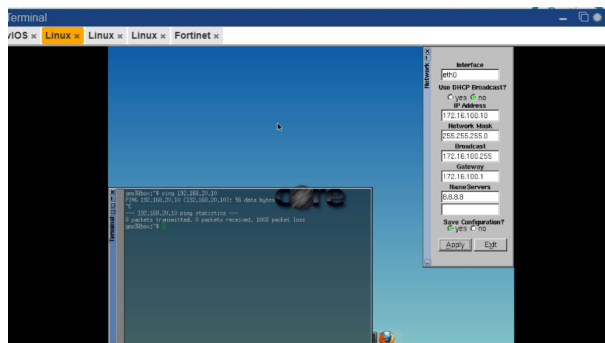
Gambar 29: Pengujian ping Client LAN ke server DMZ



Gambar 30: Pengujian akses web server DMZ dari Client LAN

3.8 Konfigurasi dan Pengujian Client WAN

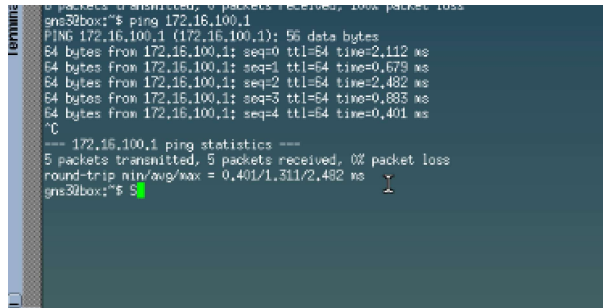
1. Mengatur IP Client WAN dengan IP 172.16.100.10/24.
2. Mengatur gateway Client WAN ke 172.16.100.1.



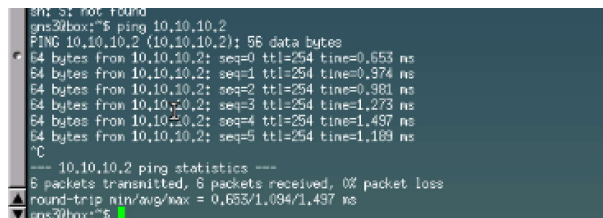
Gambar 31: Konfigurasi IP Client WAN

3. Melakukan ping dari Client WAN ke gateway MikroTik ISP.

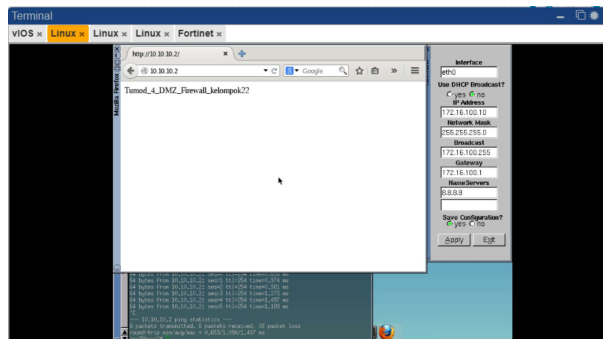
4. Melakukan ping dari Client WAN ke FortiGate.
5. Melakukan akses web server DMZ melalui VIP FortiGate.



Gambar 32: Pengujian ping Client WAN ke gateway



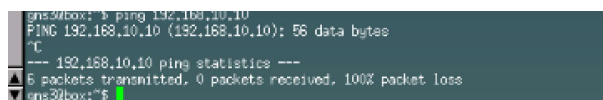
Gambar 33: Pengujian ping Client WAN ke FortiGate



Gambar 34: Pengujian akses web server DMZ dari Client WAN melalui VIP

3.9 Pengujian Keamanan Akses WAN

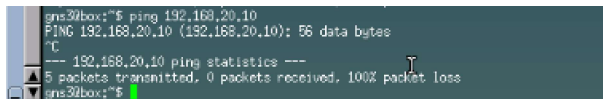
1. Melakukan ping dari Client WAN ke Client LAN.
2. Hasil ping gagal, sehingga akses langsung dari WAN ke LAN tidak diizinkan.



Gambar 35: Pengujian Client WAN tidak dapat ping ke Client LAN

3. Melakukan ping dari Client WAN ke IP asli server DMZ.

4. Hasil ping gagal, sehingga akses WAN ke DMZ hanya dilakukan melalui VIP.



Gambar 36: Pengujian Client WAN tidak dapat ping langsung ke server DMZ

3.10 Analisis Hasil Tugas Modul

Berdasarkan hasil konfigurasi dan pengujian, topologi tugas modul berhasil membagi jaringan menjadi tiga zona utama, yaitu WAN, LAN, dan DMZ. MikroTik ISP berfungsi sebagai router luar dan gateway Client WAN. Cisco Router berfungsi sebagai router internal LAN. FortiGate berfungsi sebagai firewall utama yang mengatur akses antar zona.

Client LAN dapat mengakses server DMZ melalui IP server DMZ. Client WAN juga dapat mengakses web server DMZ melalui VIP FortiGate. Pengujian keamanan menunjukkan bahwa Client WAN tidak dapat mengakses LAN secara langsung dan tidak dapat melakukan ping langsung ke IP asli server DMZ.

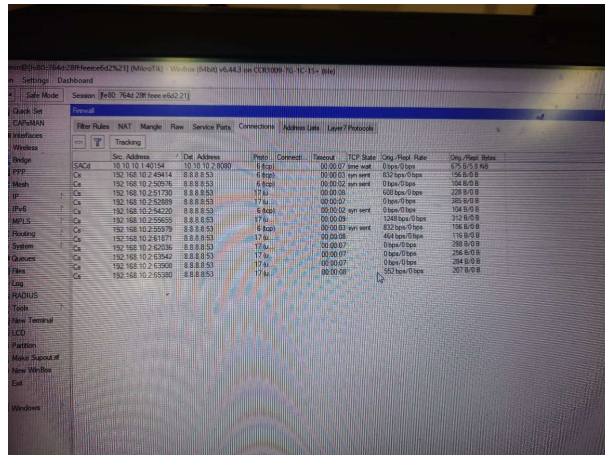
4 Kesimpulan

Dari percobaan ini dapat disimpulkan bahwa NAT dapat digunakan untuk menghubungkan jaringan lokal ke jaringan luar. Firewall filter dapat digunakan untuk membatasi akses ke router maupun akses yang melewati router. Port forwarding juga dapat digunakan untuk meneruskan akses dari jaringan luar menuju web server lokal.

Pada tugas modul, konfigurasi WAN, LAN, dan DMZ berhasil diterapkan. Client LAN dapat mengakses server DMZ, sedangkan Client WAN dapat mengakses web server DMZ melalui VIP FortiGate. Akses langsung dari WAN ke LAN dan ke IP asli DMZ dapat dibatasi menggunakan firewall policy.

5 Lampiran

5.1 Dokumentasi saat praktikum



Service	Src. Address	Dest. Address	Port	Connect	Timeout	TCP State	Orig. Rate	Orig. Rate
Bridge	10.10.10.141/24	10.10.10.10/24	8080	0	00:00:00	new wait	0 bps	0 bps
PPP	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps
Mesh	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps
IP	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps
IPS	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps
MPLS	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps
Routing	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps
System	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps
Queue	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps
Rm	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps
Log	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps
RADUS	192.168.10.2/24	192.168.10.2/24	8080	0	00:00:00	new wait	0 bps	0 bps

Gambar 37: Hasil connection tracking